

A disproof of Conjecture 00000004055

Over the self-injective algebra $k[x]/(x^2)$ every syzygy period is 1, so no prime period occurs

1 The conjecture

Conjecture 00000004055 reads (English):

Definition: The syzygy $\Omega(M)$ is the kernel of the minimal projective cover, and the period is the minimal number of iterations first returning to an isomorphic module.

Conjecture: On every self-injective algebra the syzygy period of any prescribed prime p can be realized, and the minimal period 2 is attained by open-chain modules.

The Chinese text says the same: on every self-injective algebra, a syzygy period equal to any given prime p can be realized.

Reading. Clause 1: for every self-injective algebra R and every prime p there is an R -module M of syzygy period p . Clause 2: every self-injective algebra has a module of period 2 (an “open-chain module”; this notion is undefined, and we show that no module at all has period 2 in our example).

We refute both clauses with one algebra, $A = k[\varepsilon]$ with $\varepsilon^2 = 0$ (isomorphic to $k[x]/(x^2)$), over any field k . It is a commutative 2-dimensional algebra, and it is not semisimple (the ideal εA is a nonzero nilpotent ideal).

2 Definitions

Let R be a ring.

Definition 1. A submodule $K \leq P$ is *superfluous* if $K + L = P$ implies $L = P$ for every submodule L . A *projective cover* of M is an epimorphism $\pi : P \rightarrow M$ with P projective and $\ker \pi$ superfluous in P . (R is *self-injective* if R is injective as an R -module.)

These are the standard notions: Wikipedia, *Projective cover*: “A projective cover is a pair (P, p) , with P a projective object in $\mathcal{C}$ “and p a superfluous epimorphism”, where for R -modules “a superfluous epimorphism is then an epimorphism” $p : P \rightarrow M$ “such that the kernel of p is a superfluous submodule of P ”. Wikipedia, *Essential extension*: “The dual notion of an essential submodule is that of superfluous submodule (or small submodule).” It defines them as in Definition 1. Wikipedia, *Injective module*: “If a ring is injective over itself as a right module, then it is called a right self-injective ring.”

Definition 2 (Lean: `IsSyzygy`, `IterSyz`, `HasPeriod`). N is a syzygy of M if $N \cong \ker \pi$ for some projective cover $\pi : P \rightarrow M$. $\Omega^0(M) \cong N$ means $M \cong N$, and $\Omega^{n+1}(M) \cong N$ means that there is a syzygy M' of M with $\Omega^n(M') \cong N$. M has *period* p if $p \geq 1$, $\Omega^p(M) \cong M$, and $\Omega^m(M) \not\cong M$ for $1 \leq m < p$.

Projective covers are unique up to isomorphism when they exist, so this agrees with the usual functional Ω . We do not need that fact: Theorem 6 shows that every syzygy of a periodic module is isomorphic to it, so the result holds for any choice of covers. Modules are arbitrary (not required to be finitely generated); this only weakens the existence claim.

Conventions. $\mathbb{N}$ contains 0; periods are ≥ 1 . Clause 1 is formalized for commutative algebras (`PrimePeriodsRealized`): for every commutative ring R that is a finite-dimensional k -algebra with R injective over itself, and every prime p , some R -module (in `Type`) has period p . For a commutative ring, left and right self-injectivity coincide, and the clause for all self-injective algebras implies this one.

3 Proof

Lemma 3 (`selfInjective`). *A is self-injective.*

Proof. Baer's criterion (`Mathlib Module.Baer.injective`): let I be an ideal and $g : I \rightarrow A$ linear. If I contains an element $x = a + b\varepsilon$ with $a \neq 0$, then x is a unit, $I = A$ and g extends to itself. Otherwise $I \subseteq \varepsilon A$. If $I = 0$, take 0. Otherwise $x_0 = b_0\varepsilon \in I$ with $b_0 \neq 0$; $c = g(x_0)$ satisfies $\varepsilon c = g(\varepsilon x_0) = 0$, so $c = c_1\varepsilon$, and $r \mapsto r(c_1/b_0)$ extends g . $\square$

Lemma 4 (`mem_eps_of_eps_smul`, `superfluous_le_eps`). *Let P be a projective A -module. (a) If $\varepsilon p = 0$ then $p \in \varepsilon P$. (b) Every superfluous submodule K of P lies in εP .*

Proof. P has a splitting $s : P \rightarrow A^{(P)}$ of the map $A^{(P)} \rightarrow P$ (`Mathlib Module.projective_def'`). If every coordinate of $s(p)$ lies in εA , then $s(p) = \varepsilon f$ and $p = \varepsilon \cdot (\text{image of } f) \in \varepsilon P$. (a): $\varepsilon s(p) = s(\varepsilon p) = 0$ forces every coordinate into εA . (b): if $y \in K \setminus \varepsilon P$, some coordinate $\varphi(y) = s(y)_i$ is a unit. Let $L = \{p : \varphi(p) \in \varepsilon A\}$, a submodule with $y \notin L$. Every p equals $cy + (p - cy)$ with $c = \varphi(p)_0/\varphi(y)_0$ (constant terms), and $p - cy \in L$. So $K + L = P$ but $L \neq P$, a contradiction. $\square$

Lemma 5 (`eps_smul_ker`, `ker_equiv_of_eps`). *Let $\pi : P \rightarrow M$ be a projective cover. Then ε kills $\ker \pi$. If ε kills M , then $\ker \pi \cong M$.*

Proof. By (b), $\ker \pi \subseteq \varepsilon P$, and $\varepsilon^2 = 0$. If $\varepsilon M = 0$, then $\mu : P \rightarrow \ker \pi$, $p \mapsto \varepsilon p$, is well defined and onto (by (b)), and $\ker \mu = \ker \pi$: if $\varepsilon p = 0$ then $p = \varepsilon q$ by (a), so $\pi(p) = \varepsilon \pi(q) = 0$; conversely $\ker \pi$ is killed by ε . Hence $\ker \pi \cong P/\ker \mu = P/\ker \pi \cong M$. $\square$

Theorem 6 (`periodic_imp_period_one`, `period_eq_one`, `no_prime_period`). *Let M be an A -module with $\Omega^n(M) \cong M$ for some $n \geq 1$. Then every syzygy of M is isomorphic to M , and $\Omega^1(M) \cong M$. Hence every module of finite period has period 1, and no A -module has period p for any prime p .*

Proof. Every syzygy is killed by ε (`syzygy_killed`), so $M \cong \Omega^n(M)$ is killed by ε (`iterSyz_killed`). By the lemma, every syzygy of M is isomorphic to M . If M had period $p \geq 2$, then $\Omega^1(M) \cong M$ would contradict minimality. $\square$

Theorem 7 (`simple_has_period_one`, `minimal_period_is_one`). *The simple module εA has period 1: multiplication by ε , $A \rightarrow \varepsilon A$, is a projective cover with kernel εA . So the least period over A is 1, and no module has period 2.*

Corollary 8 (`conjecture_4055_false`, `period_two_false`). *For every field k , `PrimePeriodsRealized k` is false, and so is the period-2 clause (every finite-dimensional commutative self-injective k -algebra has a module of period 2).*

4 Scope

Refuted: the literal reading (for every self-injective algebra and every prime p some module has period p), already for $p = 2$ and for every p ; and the period-2 clause, whatever “open-chain module” means. Not refuted: (a) the reading “for every prime p some self-injective algebra has a module of period p ”; (b) the reading in which $\Omega^n(M) \cong M$ is taken in the stable module category (isomorphism up to projective summands). It is not formalized and we make no claim about it, although Lemma 5 and `syzygy_killed` show that every syzygy N over A already satisfies $\Omega(N) \cong N$.

5 Lean formalization

File `lean/Conjecture4055/Basic.lean` (309 lines, Lean 4.33.1, Mathlib v4.33.1), namespace `C4055`. Main theorem: `C4055.conjecture_4055_false (k : Type) [Field k] : ~PrimePeriodsRealized k`. `#print` axioms for `conjecture_4055_false`, `period_two_false`, `no_prime_period`, `periodic_imp_period_one`, `minimal_period_is_one` and `selfInjective` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.

Sources

- https://en.wikipedia.org/wiki/Projective_cover
- https://en.wikipedia.org/wiki/Essential_extension
- https://en.wikipedia.org/wiki/Injective_module